Digital Signal Processing - Sampling, DFT & FFT

Department: Electronics & Communication | Semester: 5 | Code: EC502
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1. Nyquist-Shannon Sampling Theorem

A continuous-time signal must be sampled at a rate fs >= 2*f_max to reconstruct the original signal without aliasing distortion.

2. Discrete Fourier Transform & Radix-2 FFT

Cooley-Tukey Radix-2 FFT algorithm reduces computational complexity from O(N^2) to O(N log2 N) using divide-and-conquer twiddle factors.

3. Digital Filter Design (FIR vs IIR)

FIR filters are inherently stable with linear phase characteristics; IIR filters achieve sharper cutoff responses with lower order polynomials.